

LP2 (P3- KRUSKAL'S MINIMAL SPANNING TREE)

Implement Greedy search algorithm for any of the following application:
Kruskal's Minimal Spanning Tree Algorithm

```
def find(parent, i):
    while parent[i] != i:
        i = parent[i]
    return i

def union(parent, x, y):
    parent[x] = y

v = int(input("Enter number of vertices: "))
e = int(input("Enter number of edges: "))
edges = []
print("Enter edges (u v w):")

for i in range(e):
    u, v1, w = map(int, input().split())
    edges.append([w, u, v1])
edges.sort()
parent = [i for i in range(v + 1)] # FIX
cost = 0
print("\nEdges in MST:\n")

for w, u, v1 in edges:
    x = find(parent, u)
    y = find(parent, v1)
```

```
if x != y:
    print(u, "-", v1, "=", w)
    cost += w
    union(parent, x, y)
print("\nMinimum Cost =", cost)
```

Kruskal's Algorithm – Viva Questions

1. What is Kruskal's Algorithm?

It is a greedy algorithm used to find the **Minimum Spanning Tree (MST)** of a connected weighted graph.

2. What is a Minimum Spanning Tree (MST)?

A tree that:

- Connects all vertices
 - Has no cycles
 - Has minimum total edge weight
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3. Which algorithm technique is used in Kruskal's?

 Greedy technique

A **Greedy Search Algorithm** is a technique where we:

Always choose the **best option available at the current step** without thinking about future consequences.

4. Why is it called greedy?

Because it always picks the **minimum weight edge first**.

5. What are vertices?

Vertices are nodes in a graph.

6. What are edges?

Edges are connections between vertices with a weight.

7. What is the format of input in Kruskal?

$u \ v \ w$

Where:

- u = source vertex
 - v = destination vertex
 - w = weight
-

8. What is Union-Find / Disjoint Set?

A data structure used to detect cycles in Kruskal's algorithm.

9. Why do we use Union-Find?

To check whether adding an edge forms a cycle or not.

10. What is a cycle?

A path that starts and ends at the same vertex.

11. What is sorting used for in Kruskal?

Edges are sorted in **increasing order of weight**.

12. Why do we sort edges?

To always pick the smallest edge first (greedy choice).

13. Time complexity of Kruskal's Algorithm?

$O(E \log E)$

14. What is E in complexity?

E = number of edges

15. What is the difference between Kruskal and Prim?

Kruskal**Prim**

Edge-based

Vertex-based

Works on sorted edges Works from a starting node

Uses Union-Find

Uses priority queue

16. Does Kruskal work on disconnected graphs?

Yes, it creates a **Minimum Spanning Forest**.

17. What is greedy choice property?

Choosing the best local option at each step.

18. What is optimal substructure?

Optimal solution can be built from optimal sub-solutions.

19. Why do we avoid cycles?

Because MST must be a tree (no cycles allowed).

20. What is Union operation?

It combines two sets into one.

21. What is Find operation?

It finds the root parent of a node.

22. What is path compression?

A technique to make Find operation faster by directly linking nodes to root.

23. Why is Kruskal called a greedy algorithm?

Because it selects edges based only on minimum weight without reconsidering past decisions.

24. What data structure is used in your program?

- List (edges)
 - Disjoint Set (Union-Find)
-

25. What is MST used for in real life?

- Network design (LAN, internet)
 - Road construction
 - Electrical wiring
 - Pipeline layout
-

26. Why do we not include all edges?

Because including all edges creates cycles and increases cost.

27. What is the condition to include an edge?

If it does **not form a cycle**.

28. What is the output of Kruskal's algorithm?

- Selected MST edges
 - Total minimum cost
-

29. What is a weighted graph?

A graph where each edge has a cost/weight.

30. One-line definition of Kruskal?

"Kruskal's algorithm finds the Minimum Spanning Tree by selecting edges in increasing order of weight while avoiding cycles."